Proxmox VE Migration

Day 3: Hardening, Operations & Production Cutover

Day 3 — Agenda

1. Recap: Day 2 Key Points
2. Security & Hardening for the Migrated Platform
3. Host OS Hardening (Debian Baseline)
4. Identity, RBAC & API Tokens
5. Network Segmentation & Firewall
6. Storage Security & Encryption
7. Monitoring & Logging Integration

8. Backup & Disaster Recovery with PBS
9. PCIe Passthrough (GPUs) Considerations
10. Cluster Basics & High Availability
11. Shared Storage & Ceph Outlook
12. Production Best Practices & Cutover Checklist
13. Next Steps & Course Wrap-up

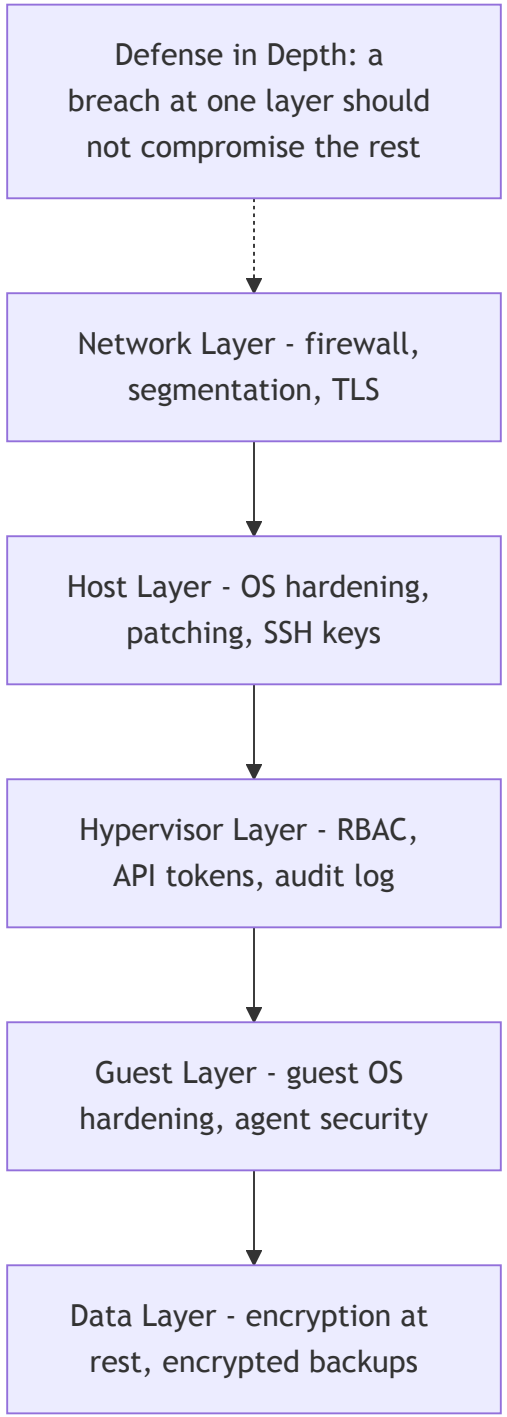
1. Recap: Day 2 Key Points

- ESXi import is a **cold migration**; keep source VMs untouched
- Linux is usually straightforward; Windows needs **VirtIO drivers**
- Guest adaptation covers **agent, NIC, disk, firmware, time sync**
- A written **function-test matrix** defines "done"
- Troubleshoot from task logs and `qm config`; change one thing at a time
- Capture a **performance baseline** before and after migration

Today we make the migrated platform **secure, observable, backed up, and production-ready**.

2. Security & Hardening for the Migrated Platform

Security must be designed into the target, not bolted on after cutover. Migration is a good moment to reset accumulated VMware technical debt.


Defense in Depth: a breach at one layer should not compromise the rest
Network Layer - firewall, segmentation, TLS
Host Layer - OS hardening, patching, SSH keys
Hypervisor Layer - RBAC, API tokens, audit log
Guest Layer - guest OS hardening, agent security
Data Layer - encryption at rest, encrypted backups

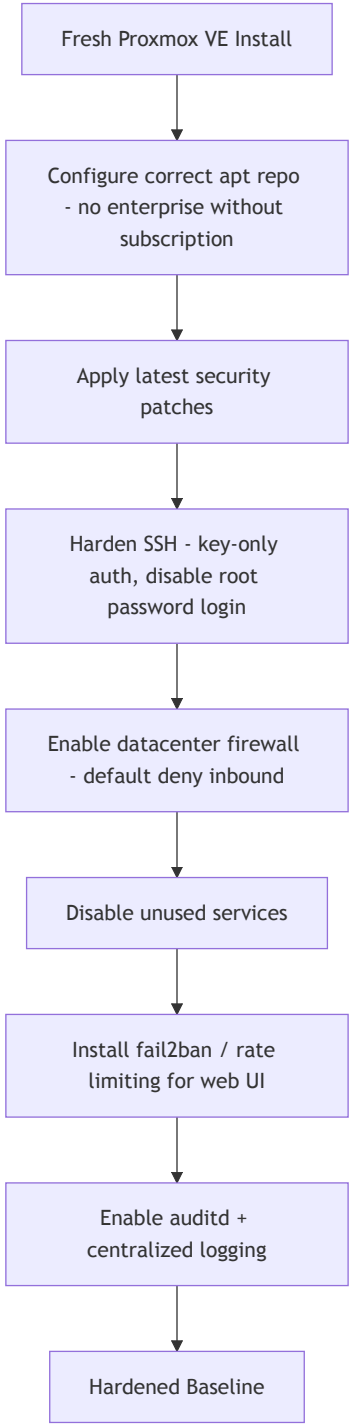
Threat actors to design against:

- External attackers targeting web UI/API (port 8006) or SSH
- Malicious or compromised insiders with valid credentials
- VM/container breakout from a migrated, possibly unpatched guest
- Supply-chain and backup-target compromise

Layered approach: host hardening → identity → network segmentation → storage protection → logging/monitoring → backup/DR.

3. Host OS Hardening (Debian Baseline)

Proxmox VE runs on Debian — general Linux hardening applies, plus Proxmox-specific considerations.


Fresh Proxmox VE Install
Configure correct apt repo - no enterprise without subscription
Apply latest security patches
Harden SSH - key-only auth, disable root password login
Enable datacenter firewall - default deny inbound
Disable unused services
Install fail2ban / rate limiting for web UI
Enable auditd + centralized logging
Hardened Baseline

```
# SSH: key-only auth, no root password login
sed -i 's/^#PermitRootLogin.*/PermitRootLogin prohibit-password/' /etc/ssh/sshd_config
sed -i 's/^#PasswordAuthentication.*/PasswordAuthentication no/' /etc/ssh/sshd_config
systemctl restart sshd

# Keep only the repositories you actually need
rm -f /etc/apt/sources.list.d/pve-enterprise.list  # if not licensed
apt update && apt full-upgrade -y

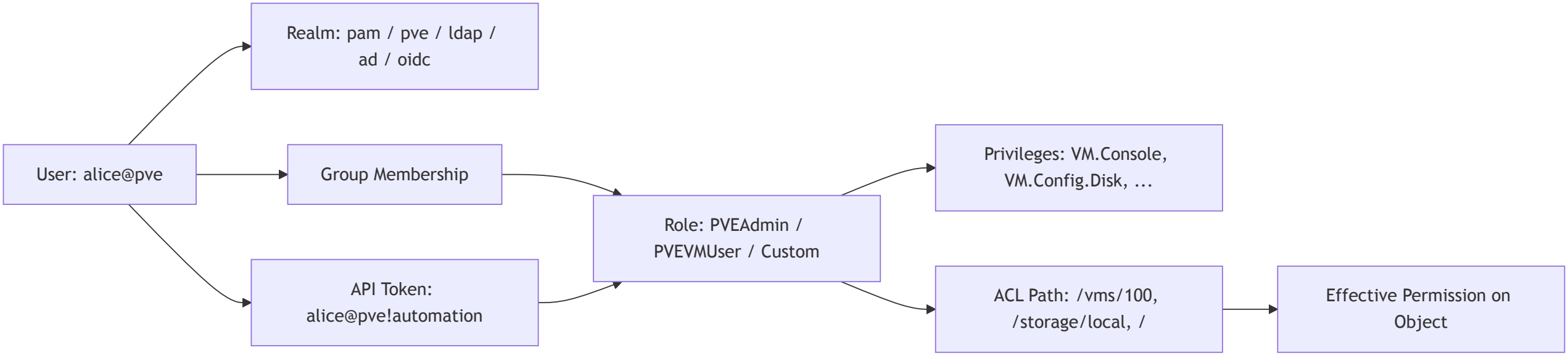
# Time sync
timedatectl set-ntp true
```

Host hardening rules:

- Avoid installing unrelated packages/services on hypervisor hosts
- Restrict SSH to a bastion/admin jump host where possible
- Use separate management and VM networks
- Keep firmware/BMC and OS patched on a defined cadence
- Store host configs in version control (Ansible/Terraform) instead of ad-hoc edits

4. Identity, RBAC & API Tokens

Proxmox VE uses **Users → Realms → Groups → Roles → ACL paths**. Migrate from shared `root` passwords to scoped, auditable access.


Realm: pam / pve / ldap / ad / oidc
User: alice@pve
Group Membership
API Token: alice@pve!automation
Role: PVEAdmin / PVEVMUser / Custom
Privileges: VM.Console, VM.Config.Disk, ...
ACL Path: /vms/100, /storage/local, /
Effective Permission on Object

```
# Realm + user + scoped role
pveum realm add ad-corp --type ad --server ad.example.com --domain example.com
pveum user add jsmith@ad-corp
pveum group add pve-admins
pveum user modify jsmith@ad-corp --groups pve-admins
pveum acl modify / --groups pve-admins --roles Administrator

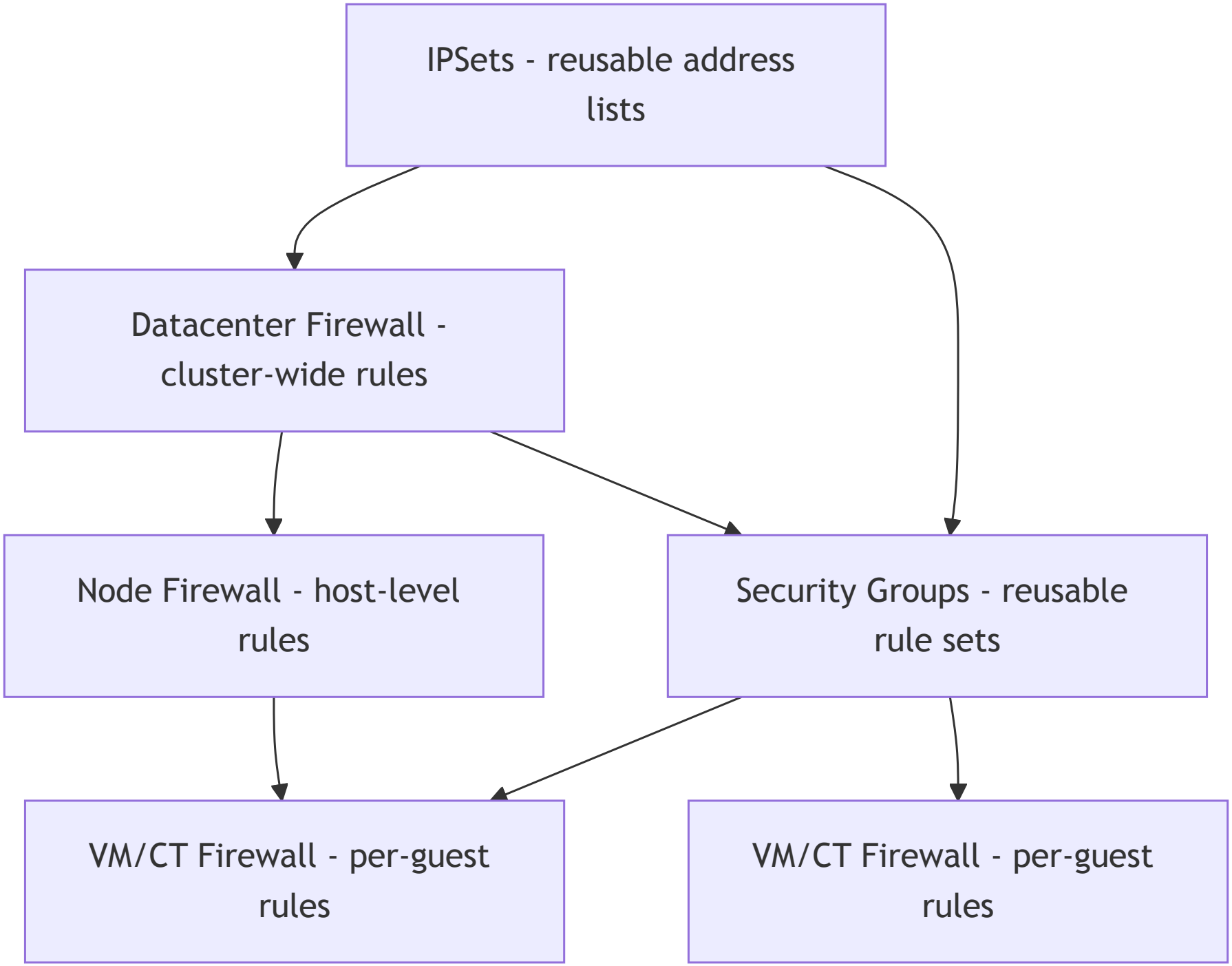
# API token for automation (Terraform/Ansible/CI)
pveum user add automation@pve
pveum user token add automation@pve terraform --privsep 1
```

Best practices:

- Enforce **2FA** (TOTP/U2F/WebAuthn) for all human admin accounts
- Use **API tokens** with privilege separation for automation; never embed root passwords
- Grant least privilege per path (`/vms/100` , `/nodes/pve1` , `/storage/backup`)
- Review ACLs quarterly and after staff changes
- Prefer directory-backed realms (AD/LDAP/OIDC) for centralized offboarding

5. Network Segmentation & Firewall

Proxmox VE has a built-in firewall at three levels: **datacenter, node, and VM/container**. Enable it even for "internal" clusters.


IPSets - reusable address lists
Datacenter Firewall - cluster-wide rules
Node Firewall - host-level rules
Security Groups - reusable rule sets
VM/CT Firewall - per-guest rules
VM/CT Firewall - per-guest rules

```
# Enable firewall at datacenter level (web UI: Datacenter -> Firewall -> Options)
# Node-level firewall also has its own enable switch
# Per-VM firewall is enabled in the VM's firewall tab
```

Segmentation plan:

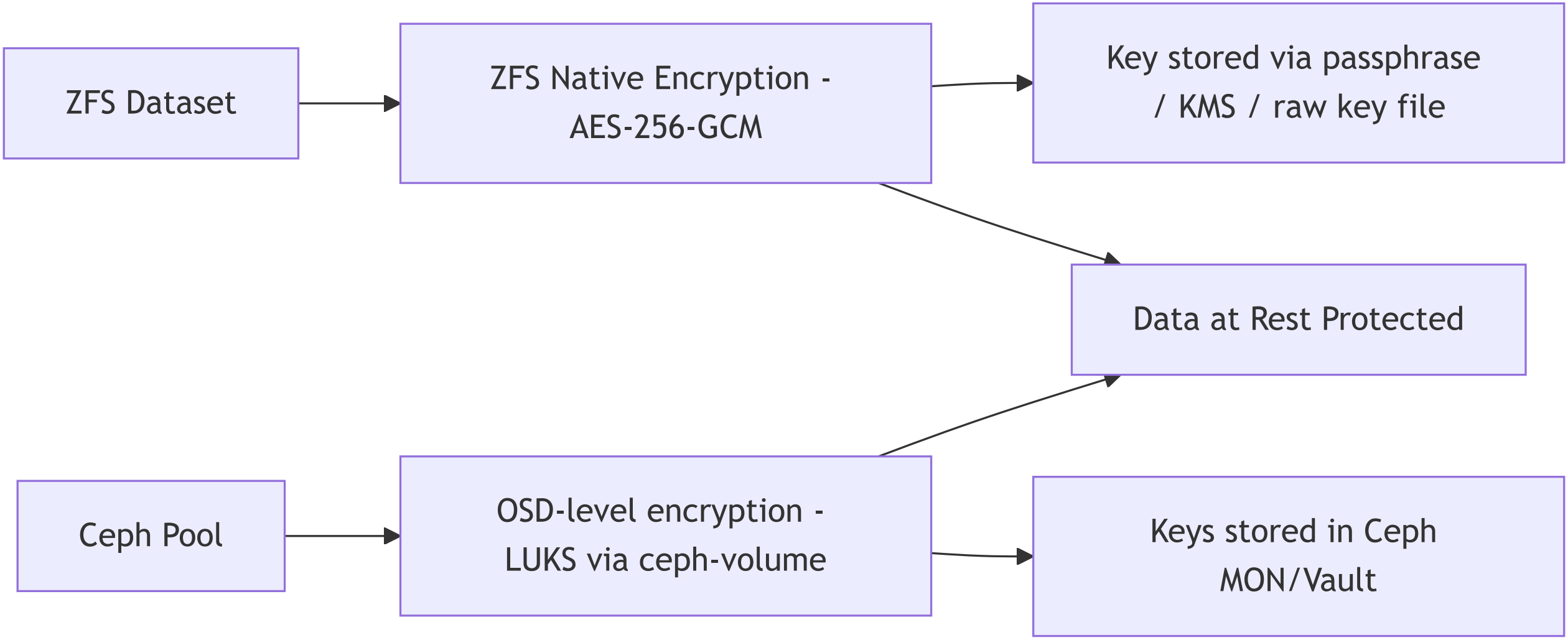
Zone	Examples	Controls
Management	Proxmox UI/API, SSH, SSH bastion	Restrict to admin IPs/VPN
Migration	Source ESXi, transfer host, target nodes	Temporary, remove after cutover
Production VMs	App/DB tiers	Micro-segmentation rules
Backup	PBS, backup targets	Only backup traffic
Storage	NFS/Ceph/iSCSI	Only storage nodes

Rules of thumb:

- Default-deny between zones; allow only required ports
- Do not expose port 8006 to the public internet without a VPN/reverse proxy + WAF
- Use VLAN-aware bridges or SDN for multi-tenant isolation
- Document every firewall change in version control

6. Storage Security & Encryption

Protect data at rest and in transit, especially after moving off VMware's familiar stack.


ZFS Dataset
ZFS Native Encryption - AES-256-GCM
Key stored via passphrase / KMS / raw key file
Data at Rest Protected
Ceph Pool
OSD-level encryption - LUKS via ceph-volume
Keys stored in Ceph MON/Vault

Options:

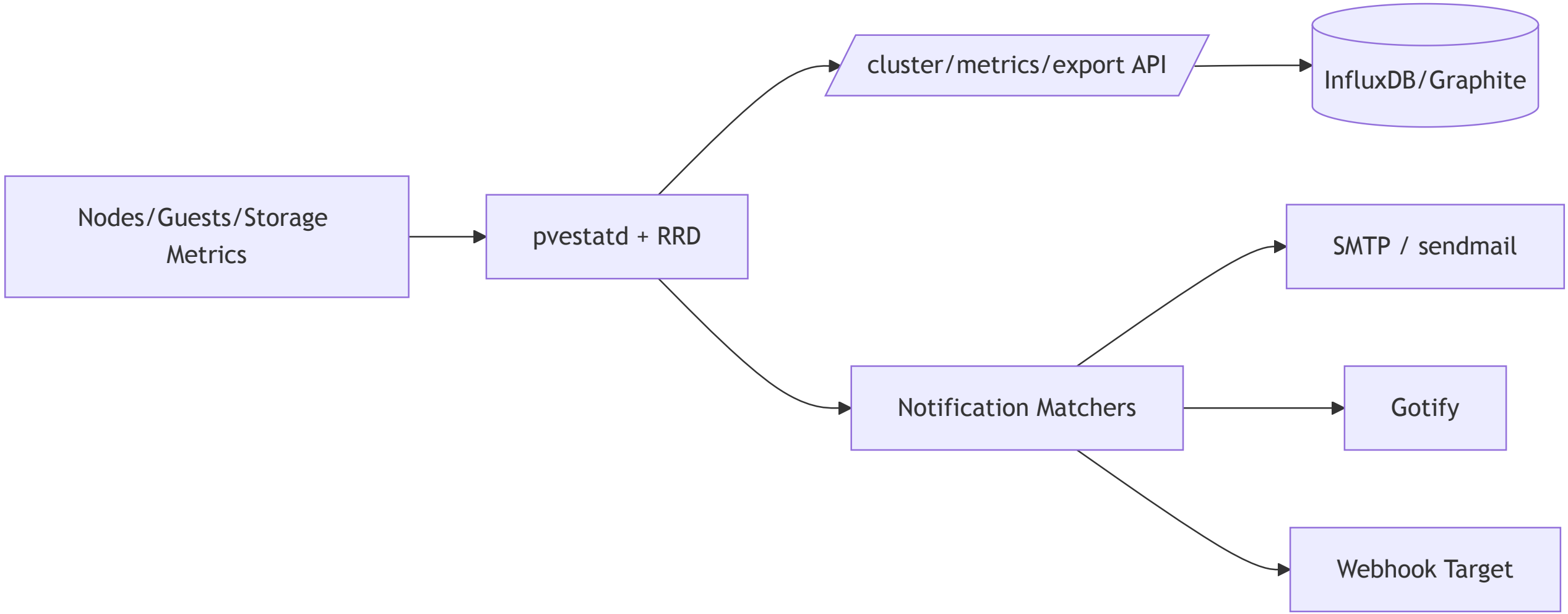
Layer	Mechanism	Notes
Disk	LUKS (Linux)	Host-level encryption; key management burden
ZFS	Native encryption (dataset-level)	Good balance; careful with key storage
Ceph	OSD-level encryption	For hyper-converged clusters
VM disk	Guest OS encryption (BitLocker, LUKS)	Strongest end-to-end; guest-managed

Layer	Mechanism	Notes
Backup	PBS encryption (client-side)	Recommended; keys must be backed up separately
Transport	TLS for PBS, Ceph, NFS over VPN	Encrypt management/storage networks

Key management warning: encryption is only as good as key recovery. Store keys in a vault (Vault, 1Password, HSM) and test restore of an encrypted backup *before* relying on it.

7. Monitoring & Logging Integration

You cannot operate what you cannot see. Migrate monitoring along with (or before) workloads.


Nodes/Guests/Storage Metrics
pvestatd + RRD
cluster/metrics/export API
InfluxDB/Graphite
Notification Matchers
SMTP / sendmail
Gotify
Webhook Target

Proxmox built-ins:

- `pvestatd` collects CPU/RAM/disk/network metrics into RRD (visible in UI graphs)
- Notification matchers for email, Gotify, and webhooks
- Task logs for every operation (import, backup, migration)

External integration:

```
# Export metrics to InfluxDB/Graphite
# /etc/pve/status.cfg
influxdb: influx-prod
    server 10.10.10.50
    port 8089
    # ...
```

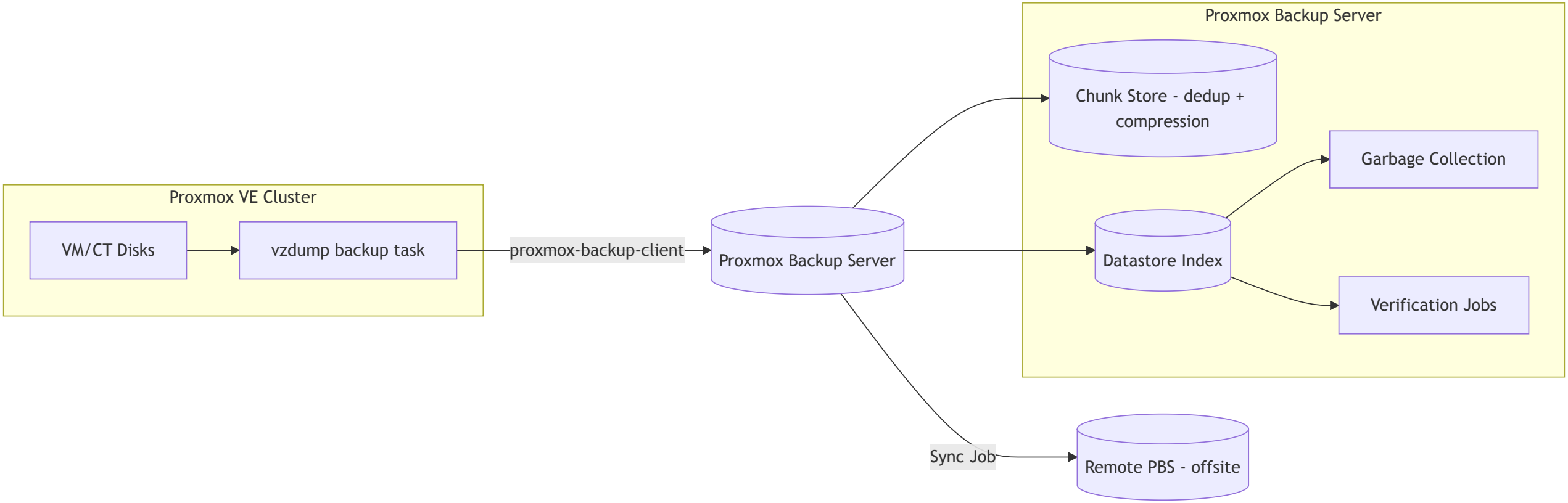
Logging:

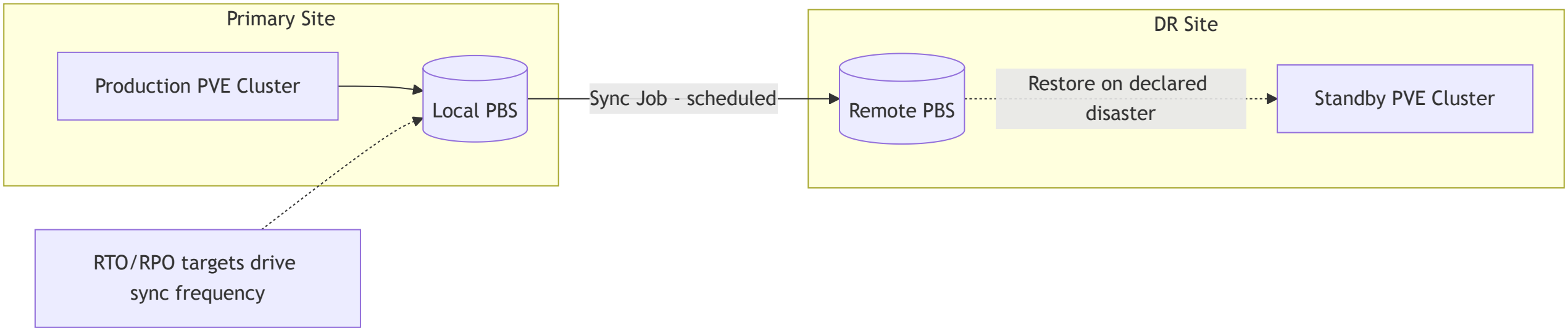
- Forward syslog from nodes and guests to a central SIEM (rsyslog/syslog-ng)
- Retain Proxmox task logs and PBS logs per compliance policy
- Monitor backup success/failure, not just "job ran"

Suggested dashboards: node health, storage capacity/IOPS, backup success rate, VM power state changes, firewall drops, certificate expiry.

8. Backup & Disaster Recovery with PBS

Proxmox Backup Server (PBS) is the natural backup target for a Proxmox estate: deduplicated, incremental, encrypted, with file-level restore.


Proxmox Backup Server
Chunk Store - dedup + compression
Garbage Collection
Proxmox VE Cluster
VM/CT Disks
vzdump backup task
proxmox-backup-client
Proxmox Backup Server
Datastore Index
Verification Jobs
Sync Job
Remote PBS - offsite


Primary Site
Production PVE Cluster
Local PBS
Sync Job - scheduled
DR Site
Remote PBS
Restore on declared disaster
Standby PVE Cluster
RTO/RPO targets drive sync frequency

Backup design:

Setting	Recommendation
Target	PBS on separate hardware/site
Schedule	Daily incremental, weekly full (or PBS's incremental-forever)
Retention	7 daily / 4 weekly / 12 monthly (example)
Encryption	Client-side, keys in vault
Verification	Scheduled verify jobs + periodic restore tests
Off-site	Replicate PBS datastore to a second site

```
# Add PBS storage on Proxmox
pvesm add pbs pbs-prod --server pbs.example.com --datastore main \
  --username backup@pbs --password '***' --fingerprint '...'
```

DR rule: a backup you have never restored is a wish, not a backup. Schedule quarterly restore tests of representative VMs.

9. PCIe Passthrough (GPUs) Considerations

GPU/PCIe passthrough is often the reason a workload stayed on VMware. It is supported on Proxmox but needs planning.

Requirements:

- CPU and motherboard with **IOMMU** (Intel VT-d / AMD-Vi) enabled in BIOS
- Kernel parameters: `intel_iommu=on` or `amd_iommu=on` (and often `iommu=pt`)
- VFIO modules loaded; device isolated in its own IOMMU group (ACS may be required)
- VM configured with `hostpci0: 0000:xx:xx.x` and a Q35 machine type

```
# /etc/default/grub
GRUB_CMDLINE_LINUX_DEFAULT="quiet intel_iommu=on iommu=pt"
update-grub
reboot

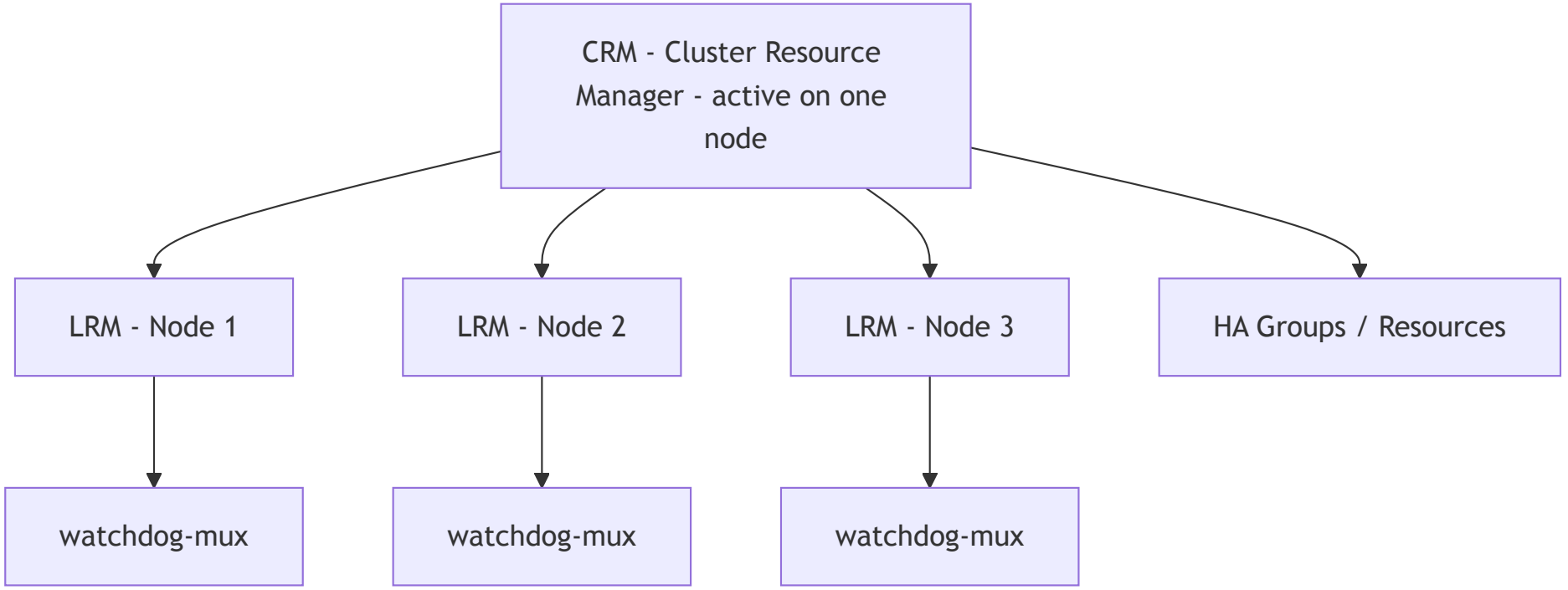
# Verify IOMMU groups
find /sys/kernel/iommu_groups/ -type l | head
```

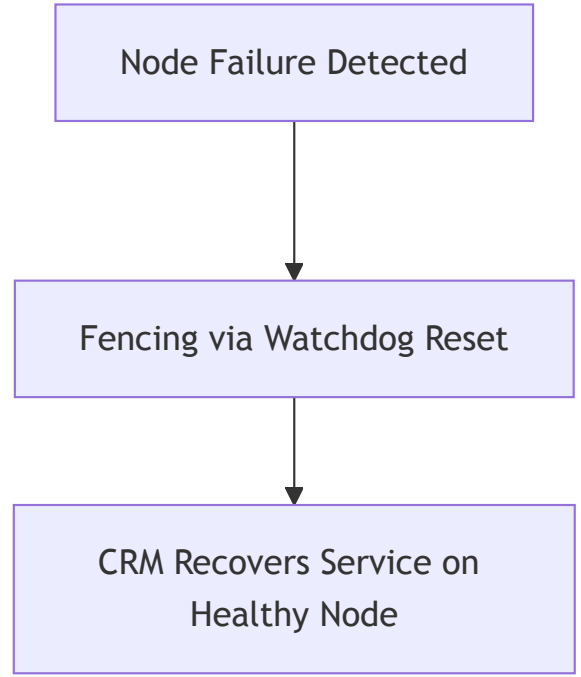
Gotchas:

- Some GPUs (especially consumer NVIDIA) fail to reset between VM restarts ("code 43")
- vGPU (mediated devices) requires vendor licensing and supported hardware
- Passthrough VMs generally cannot be live-migrated
- HA restart is possible but the device must reinitialize cleanly
- Power/thermal: GPUs in virtualized hosts still need physical cooling and power budgeting

10. Cluster Basics & High Availability

Once more than one node exists, you have a cluster. Clusters unlock shared config, live migration, and HA.


CRM - Cluster Resource Manager - active on one node
LRM - Node 1
LRM - Node 2
LRM - Node 3
HA Groups / Resources
watchdog-mux
watchdog-mux
watchdog-mux


Node Failure Detected
Fencing via Watchdog Reset
CRM Recovers Service on Healthy Node

```
# Create cluster on first node
pvecm create prod-cluster

# Join additional nodes
pvecm add pve1.example.com

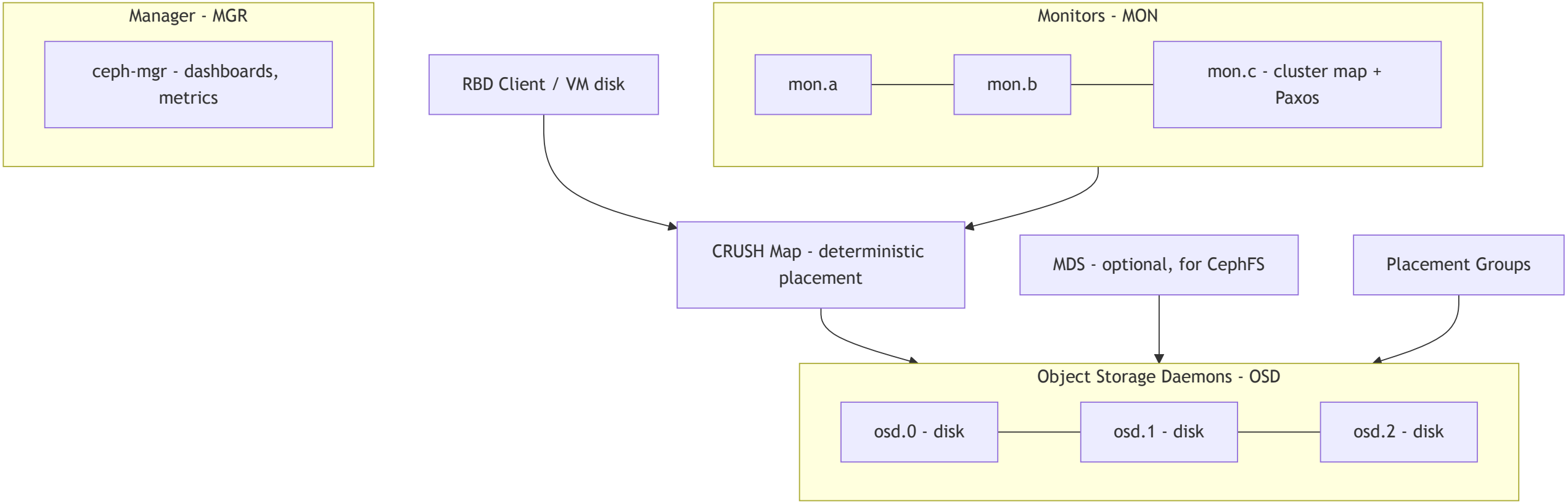
# HA: add a VM to HA management
ha-manager add vm:100
```

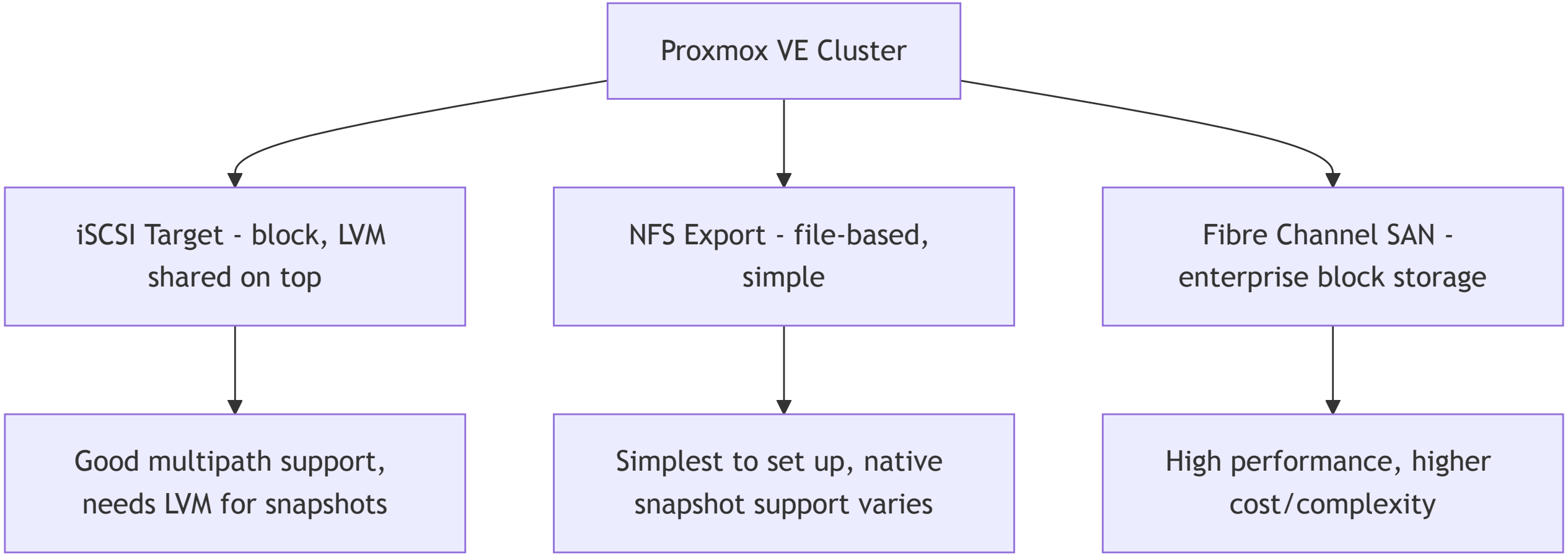
HA rules:

- Requires shared storage (or replicated storage) for the VM disks
- Requires quorum (majority of votes) — use a QDevice for even-node clusters
- HA restarts VMs on another node; it is **not** continuous availability
- Fencing (watchdog) prevents split-brain; test it deliberately in a lab
- Start with non-critical VMs in HA; expand after confidence

11. Shared Storage & Ceph Outlook

For live migration and HA you need shared storage. Proxmox supports NFS/iSCSI/SAN and hyper-converged Ceph.


Manager - MGR
ceph-mgr - dashboards, metrics
RBD Client / VM disk
Monitors - MON
mon.a
mon.b
mon.c - cluster map + Paxos
CRUSH Map - deterministic placement
MDS - optional, for CephFS
Placement Groups
Object Storage Daemons - OSD
osd.0 - disk
osd.1 - disk
osd.2 - disk


Proxmox VE Cluster
iSCSI Target - block, LVM shared on top
NFS Export - file-based, simple
Fibre Channel SAN - enterprise block storage
Good multipath support, needs LVM for snapshots
Simplest to set up, native snapshot support varies
High performance, higher cost/complexity

Approach	Pros	Cons
NFS (existing NAS)	Simple, reuses investment	Single point of failure unless HA NAS
iSCSI/SAN	Block storage, familiar to VMware admins	Licensing, complexity, network design
Ceph (hyper-converged)	Scale-out, self-healing, integrated	Needs 3+ nodes, 10GbE+, operational skill
ZFS replication	Simple, no shared storage needed	Async, RPO > 0, manual failover

Ceph outlook: a great target for a later "Architect" engagement. Do not introduce Ceph and migration in the same project unless the team already has Ceph skills — keep migration storage simple first.

12. Production Best Practices & Cutover Checklist

Production best practices:

- Standardize on **templates** and **cloud-init** for new VMs
- Manage configuration with **Terraform/Ansible**; no snowflake nodes
- Patch on a defined cadence; use the enterprise repo if subscribed
- Keep at least **one off-node backup target** (PBS preferred)

- Enable firewall at datacenter level; default-deny between zones
- Monitor backup success, storage capacity, and node health
- Document network (bridges/VLANs) and storage plans
- Test HA failover and restore in a lab before relying on them
- Define RPO/RTO and verify backups meet them

Cutover checklist (per wave):

- [ ] All VMs in wave imported and adapted
- [ ] Function tests passed and recorded
- [ ] Performance baseline comparison acceptable
- [ ] Backup and snapshot tested on migrated VMs
- [ ] Monitoring and logging integrated

- [ ] Firewall rules applied and verified
- [ ] DNS/IP cutover plan rehearsed
- [ ] Rollback tested and time-boxed
- [ ] Stakeholders notified of maintenance window
- [ ] Post-cutover observation period defined (e.g., 2 weeks)
- [ ] Source VMs retained but powered off for rollback window
- [ ] Documentation and runbook updated

14. Compliance & Audit Readiness

Map Proxmox controls to your compliance framework (ISO 27001, CIS, SOC 2, BSI).

Control area	Proxmox evidence
Access control	Realms, RBAC, 2FA, API tokens, audit logs
Change management	Task logs, Git-managed configs, CAB
Backup/DR	PBS jobs, verify jobs, restore tests
Monitoring	Metrics, alerts, SIEM forwarding
Hardening	Host checklist, firewall rules
Encryption	ZFS/PBS/guest encryption
Patching	Update logs, repo policy

Compliance Evidence Collection

- Export user/ACL lists regularly
- Store firewall rules in version control
- Keep backup/restore test records
- Retain logs per policy (e.g., 12 months)
- Document risk acceptance and exceptions
- Schedule internal audits quarterly

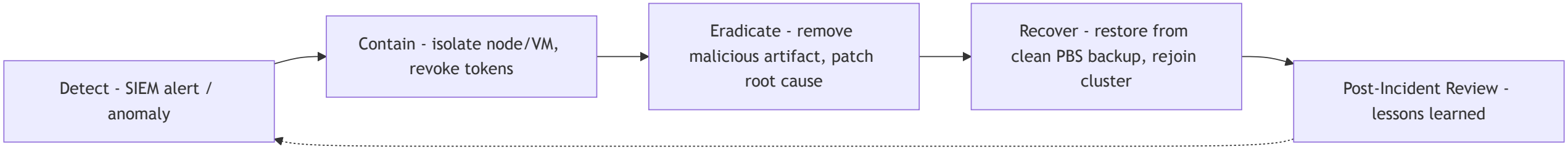
15. Advanced Host Hardening

- Enable only required services
- Disable root SSH password, enforce keys
- Configure AIDE/Tripwire for file integrity
- Enable auditd for privileged actions
- Restrict cron/at to admins
- Set umask 027 for new files
- Disable USB storage if not needed
- Configure automatic security updates (unattended-upgrades)

Kernel & Boot Hardening

```
# /etc/sysctl.d/99-hardening.conf
net.ipv4.conf.all.rp_filter=1
net.ipv4.conf.all.accept_redirects=0
net.ipv4.conf.all.send_redirects=0
net.ipv4.tcp_syncookies=1
kernel.randomize_va_space=2
fs.suid_dumpable=0
```

Apply with `sysctl --system`. Test in lab before production.


Detect - SIEM alert / anomaly
Contain - isolate node/VM, revoke tokens
Eradicate - remove malicious artifact, patch root cause
Recover - restore from clean PBS backup, rejoin cluster
Post-Incident Review - lessons learned

16. Incident Response for Virtualization

Playbook steps:

1. Detect (monitoring/alerts)
2. Triage (severity, scope)
3. Contain (isolate node/VM, firewall)
4. Eradicate (patch, remove malware)
5. Recover (restore from backup, rebuild)
6. Lessons learned (postmortem)

Incident Response — Proxmox Specifics

- Isolate a VM by disabling its network in firewall or unplugging bridge
- Snapshot before forensics (but preserve evidence)
- Use PBS to restore to isolated network for analysis
- Review task logs and syslog for timeline
- Rotate credentials if compromise suspected
- Keep a spare node for rapid rebuild

17. Logging & SIEM Integration

Forward logs from nodes and guests:

```
# rsyslog forwarding example
*.* @@siem.example.com:514
```

Collect:

- Proxmox task logs
- Authentication events
- Firewall drops
- Backup success/failure
- Storage errors
- Guest OS logs

SIEM use cases: brute-force detection, privilege escalation, anomalous migration, backup failure, capacity alerts.

Log Retention & Privacy

- Define retention per log type (e.g., 90 days hot, 12 months cold)
- Mask/anonymize PII in logs where required
- Encrypt log transport (TLS)
- Restrict log access (least privilege)
- Test log restore and search

18. Backup & Restore Drills

Quarterly drill:

1. Select 3 representative VMs (Linux, Windows, DB)
2. Restore from PBS to isolated network
3. Boot and run function tests
4. Measure restore time vs. RTO
5. Document results and gaps

Annual drill:

- Full site recovery simulation
- Include identity, DNS, monitoring, backup server itself

Backup Verification

```
# PBS verify job (schedule in PBS UI)
# proxmox-backup-manager verify <datastore>

# Check backup status from Proxmox
pvesh get /nodes/pve1/storage/pbs-prod/content
```

- Enable scheduled verify jobs
- Alert on verification failure
- Test file-level restore, not just VM restore

19. High Availability Testing

HA test plan:

1. Pick a non-critical VM in HA
2. Power off the host node (simulate failure)
3. Measure time to restart on another node
4. Verify application recovery
5. Re-add node and verify cluster health
6. Document RTO achieved

Repeat after any cluster change.

Fencing & Watchdog

- Configure watchdog (iTCO/ipmi) per node
- Test fencing in lab before production
- Monitor watchdog resets
- Understand split-brain prevention
- Keep fencing network independent

```
# Check watchdog
wdctl
systemctl status watchdog
```

20. Capacity Management

Track:

- CPU utilization and ready time
- Memory usage and ballooning
- Storage used/free and growth rate
- Network throughput and errors
- Backup capacity and dedup ratio

Forecast: 6–12 months ahead; trigger procurement early.

Capacity Alerts (Example)

Metric	Warning	Critical
Node CPU	70%	85%
Node RAM	75%	90%
Storage used	70%	85%
Backup capacity	60%	75%
HA failover count	1/week	3/week

21. Patch Management

- Define patch cadence (monthly security, quarterly feature)
- Use enterprise repo if subscribed; no-subscription otherwise
- Test patches in lab cluster first
- Patch nodes one at a time; migrate VMs off first
- Document rollback (apt history, snapshots)
- Patch guests with same discipline

```
apt update
apt list --upgradable
apt full-upgrade -y
pveversion -v
```

Patch Window Runbook

1. Announce window
2. Backup PBS + config export
3. Migrate VMs off node 1
4. Patch and reboot node 1
5. Verify node 1 health
6. Migrate VMs back (optional)
7. Repeat for each node
8. Post-window verification and comms

22. Configuration as Code

Manage Proxmox with IaC:

- Terraform: VMs, containers, users, pools
- Ansible: node config, packages, firewall, monitoring
- Git: all configs, runbooks, diagrams

Benefits: reproducibility, review, rollback, disaster recovery.

Ansible Example (Node Baseline)

```yaml
- name: Proxmox node baseline
  hosts: proxmox
  tasks:
    - name: Ensure qemu-guest-agent on guests
      # (via pct/qm exec or guest-side playbooks)
    - name: Configure NTP
      ansible.builtin.lineinfile:
        path: /etc/systemd/timesyncd.conf
        regexp: '^NTP='
        line: 'NTP=ntp.example.com'
```

23. Knowledge Transfer & Training

- Train operations team on Proxmox UI/CLI
- Create quick-reference cards
- Record short videos for common tasks
- Pair VMware admins with Linux admins during migration
- Schedule a 30-day and 90-day review
- Maintain a FAQ and troubleshooting wiki

24. Migration Retrospective

Post-migration review (2–4 weeks after last wave):

- What went well?
- What took longer than expected?
- Which runbook steps need updating?
- Performance vs. baseline results
- Incident review
- Cost vs. plan
- Next improvements (automation, HA, Ceph)

Document and share with stakeholders.

25. Appendix A — Security Hardening Checklist

- [ ] SSH key-only, no root password
- [ ] 2FA enforced for admins
- [ ] Firewall enabled at all levels
- [ ] API tokens scoped and rotated
- [ ] Hosts patched and minimal packages
- [ ] Encryption keys in vault
- [ ] Backups encrypted and tested
- [ ] Monitoring and SIEM integrated
- [ ] Break-glass account tested
- [ ] Documentation complete

Appendix B — Monitoring Dashboard List

- Node CPU/RAM/disk/network
- Storage capacity and IOPS
- VM power states and HA events
- Backup success/failure and duration
- Firewall drops
- Certificate expiry
- PBS datastore usage and dedup
- Ceph health (if used)

Appendix C — Incident Severity Matrix

Severity	Definition	Response time
Sev-1	Service down, data loss risk	15 min
Sev-2	Degraded, workaround exists	1 h
Sev-3	Minor impact	4 h
Sev-4	Cosmetic	Next business day

Appendix D — RPO/RTO Review Template

Service	RPO target	RPO achieved	RTO target	RTO achieved	Gap	Action

Review after every DR drill.

Appendix E — Cutover Communication Template

Subject: Maintenance window — VM migration to Proxmox VE

- Date/time (UTC and local)
- Affected services and users
- Expected downtime
- Rollback plan and decision point
- Contact for issues
- Post-maintenance verification steps

Send 1 week and 1 day ahead.

Appendix F — Post-Migration Optimization Ideas

- Convert frequently rebuilt VMs to templates/cloud-init
- Introduce Terraform/Ansible for new VMs
- Evaluate LXC for suitable new Linux services
- Consider HA for critical VMs
- Evaluate PBS dedup and retention tuning
- Pilot Ceph only after stability period
- Review and right-size overprovisioned VMs

Appendix G — Glossary (Migration)

Term	Meaning
Cold migration	VM powered off during import
Live migration	VM moves between nodes with minimal downtime
VirtIO	Paravirtualized drivers for KVM
pmxcfs	Proxmox cluster filesystem
PBS	Proxmox Backup Server
HA	High Availability (restart on failure)
QDevice	Tie-breaker for quorum
RPO/RTO	Recovery point/time objectives

Appendix H — Final Checklist Before Production

- [ ] All waves complete and accepted
- [ ] Backups and restore tests passed
- [ ] Monitoring and alerts live
- [ ] Firewall rules applied
- [ ] Documentation handed over
- [ ] Operations team trained
- [ ] Rollback window closed and source VMs decommissioned per policy
- [ ] Retrospective completed

26. Business Continuity Planning

- Define critical services and dependencies
- Set RPO/RTO per service
- Test failover and restore regularly
- Maintain runbooks and contact lists
- Review after major changes

27. Disaster Recovery Testing

Annual DR test:

1. Simulate site loss
2. Restore PBS and Proxmox from backup
3. Restore critical VMs
4. Verify authentication, DNS, monitoring
5. Measure RTO vs. target
6. Document gaps and improvements

28. Cost Optimization After Migration

- Right-size VMs based on actual usage
- Use templates to avoid overprovisioning
- Evaluate storage tiers (SSD vs. HDD)
- Optimize backup retention and dedup
- Review subscription needs (support vs. no-subscription)
- Decommission unused VMs

29. Green IT Considerations

- Consolidate workloads to reduce hardware
- Use power-efficient CPUs
- Enable CPU power management
- Schedule non-critical workloads off-peak
- Monitor power usage (IPMI/PDU)
- Recycle old hardware responsibly

30. Scaling Roadmap

Phase	Focus	Trigger
Stabilize	Ops, monitoring, backups	0–3 months
Optimize	Right-sizing, automation	3–6 months
Expand	HA, clustering, SDN	6–12 months
Advanced	Ceph, GPU, multi-site	12+ months

31. Vendor Support & Community

- Proxmox subscription for enterprise support
- Community forum and wiki
- Third-party consultants
- Training and certification
- Vendor appliances compatibility lists

32. Certification & Skills

- Proxmox Certified Engineer (if available)
- Linux certifications (LFCS, RHCE)
- Ceph training for storage
- Security certifications (ISO 27001, CIS)
- Internal knowledge-sharing sessions

33. Next Projects After Migration

- Automate VM provisioning (Terraform/Ansible)
- Implement HA for critical services
- Deploy PBS with off-site replication
- Pilot SDN for multi-tenant isolation
- Evaluate Ceph for scale-out storage
- Containerize suitable new workloads

34. Final Q&A

Common questions:

- How do we handle vGPU?
- What about VMware NSX?
- Can we keep some VMs on VMware?
- How do we measure success?
- What is the rollback plan?

35. Closing Remarks

Migration is a journey, not a single event. Start small, measure everything, and iterate. Proxmox VE gives you an open, flexible platform — make it yours.

Thank you for attending!

36. Appendix: Extended Security Checklist

- [ ] Hosts hardened
- [ ] 2FA enforced
- [ ] Firewall enabled
- [ ] Tokens scoped
- [ ] Logs forwarded
- [ ] Backups encrypted
- [ ] DR tested
- [ ] Docs complete

37. Appendix: Monitoring Queries (Examples)

- Node CPU > 80% for 10 min
- Storage free < 20%
- Backup job failed
- VM unexpected power off
- HA failover occurred
- Certificate expiry < 30 days

38. Appendix: Backup Retention Example

Tier	Daily	Weekly	Monthly	Yearly
Critical	14	8	12	3
Standard	7	4	6	1
Dev/test	3	2	1	0

39. Appendix: Firewall Rule Template

Direction	Source	Destination	Port	Action	Notes
In	Admin VPN	Proxmox UI	8006	Allow	Management
In	VM net	DNS	53	Allow	Name resolution
Out	Nodes	PBS	8007	Allow	Backup

40. Appendix: Patch Compliance Report Fields

- Node name
- Current version
- Pending updates
- Last patch date
- Reboot required
- Notes

41. Appendix: Capacity Forecast Template

Resource	Current	6-month	12-month	Action
CPU (pCPU)				
RAM (GB)				
Storage (TB)				

42. Appendix: Incident Postmortem Template

- Summary
- Impact
- Timeline
- Root cause
- Detection
- Resolution
- Action items
- Lessons learned

43. Appendix: Training Feedback Form

- What was most useful?
- What was missing?
- Pace (too fast/slow)
- Hands-on labs (enough?)
- Suggestions

44. Appendix: Glossary (Operations)

Term	Meaning
PBS	Proxmox Backup Server
HA	High Availability
RPO/RTO	Recovery point/time objectives
SIEM	Security info/event mgmt
IaC	Infrastructure as Code
CAB	Change Advisory Board

45. Appendix: Final Sign-Off Sheet

Role	Name	Signature	Date
Migration lead			
VMware SME			
Proxmox SME			
Security			
App owner			

46. Appendix: Quick Reference Card (Print)

- Login: https://<node>:8006
- `qm list`, `qm config <id>`
- `pvesm status`
- `pvecm status`
- `pveum user list`
- `journalctl -u pveproxy`

47. Appendix: Emergency Contacts

Role	Name	Phone	Email
Migration lead			
VMware SME			
Network			
Storage			

48. Appendix: Change Log

Version	Date	Author	Changes
1.0			Initial release

13. Next Steps & Course Wrap-up

You now have a complete migration path: assessment, preparation, import, guest adaptation, testing, hardening, monitoring, backup, and production cutover.

Where to go from here:

- **Proxmox VE Architect** — cluster design, ZFS/Ceph deep-dive, SDN, HA/DR at scale
- **Proxmox VE Security** — hardening, RBAC deep-dive, network segmentation, compliance
- **Automation** — Terraform/Ansible for Proxmox, CI/CD for VM templates
- **Operations** — PBS tuning, monitoring dashboards, runbook refinement

Final advice: migrate in small waves, measure everything, and keep the rollback path open. The import tool is the easy part; **assessment, testing, and operations** are what make a migration succeed.

Thank you for attending — questions welcome!